# Spin-$\frac{1}{2}$ Sector Identification within the Admissible Subspace: Rank Structure of the Restricted Covariance in $\mathrm{End}(V_\rho)$

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2026

**Abstract**

Paper O28 [7] established that the per-pair covariance operator $\mathcal{C}_c$ in $\mathrm{End}(H_{\mathrm{eff}})$ has rank exactly $r_{\mathrm{eff}} = 3$ with invariant normalised eigenvalue structure $[\lambda_1 : \lambda_2 : \lambda_3] = [1 : \frac{1}{2} : \frac{1}{2}]$ across all conjugate pairs and all primes $q \in \{29, 61, 101, 151\}$. The present paper identifies the structural mechanism behind this result and carries out the sector identification required for the full O26 Criterion 5.4 [5]. We prove that $r_{\mathrm{eff}} = 3$ in $\mathrm{End}(H_{\mathrm{eff}})$ is a direct consequence of the outer products $M_j = \pi_c(v_j) \otimes \pi_{q-c}(v_j)^*$ lying in the complex symmetric subspace $\mathrm{Sym}(V_\rho, \mathbb{C}) \subset \mathrm{End}(\mathbb{C}^3)$, where $V_\rho = \mathrm{span}_{\mathbb{C}}\{\pi_c(v_j)\} \subset H_{\mathrm{eff}}$ is a two-dimensional complex subspace satisfying $\dim_{\mathbb{C}} V_\rho = 2$ by the relation $k(k+1)/2 = r_{\mathrm{eff}} = 3$ with $k = 2$. Direct restriction of $\mathcal{C}_c$ to $\mathrm{End}(V_\rho)$ yields $r_{\mathrm{eff}} = 3$ in $\mathrm{End}(\mathbb{C}^2)$; the pair constraint $c \leftrightarrow q-c$ acts anti-linearly on $V_\rho$ and forces $M_j|_{V_\rho} \in \mathrm{Sym}(2, \mathbb{C})$, a three-dimensional subspace of the four-dimensional $\mathrm{End}(\mathbb{C}^2)$. The correct operationalisation of O26 Criterion 5.4 requires the full admissible morphism $\Phi_{q,\rho} = \rho \circ \iota \circ \pi$ of O27 [6]: the anti-linear pair constraint in $H_{\mathrm{eff}}$ is conjugated under $\iota$ to complex conjugation in $\mathfrak{su}(2)$, which the spin-$\frac{1}{2}$ representation $\rho$ maps to right-multiplication by the charge-conjugation matrix $C = i\sigma_2$. This transforms the anti-linear symmetric constraint into a complex-linear involution in $\mathrm{End}(\mathbb{C}^2)$, lifting the restriction to $\mathrm{Sym}(2, \mathbb{C})$ and allowing the trajectory $\{\Phi_{q,\rho}(v_c^{(n)}) \otimes \Phi_{q,\rho}(v_{q-c}^{(n)})^*\}$ to span all four dimensions of $\mathrm{End}(\mathbb{C}^2)$. The structural prediction is $r_{\mathrm{eff}} = 4 = d_\rho^2$ under the full morphism [H2], resolving O26 Criterion 5.4 and confirming the spin-$\frac{1}{2}$ sector candidate. The numerical computation from the Q5a-O5 checkpoints is presented for $q \in \{61, 101, 151\}$.

**Keywords.** spectral admissibility; Weil representation; Heisenberg graphs; covariance operator; effective dimension; spin-$\frac{1}{2}$ sector; admissible morphism; charge conjugation; Born–Infeld parity; symmetric outer products

# 1 Position of the Problem

## 1.1 What O25–O28 established

Paper O25 [4] confirmed that $\delta_{\text{pair}}$ is a structural invariant of the Weil representation of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, with inter-pair variance decreasing monotonically with $q$. Paper O26 [5] established a dictionary between conjugate Weil blocks and rank-one matrix coefficients in a representation space $V_\rho$, and formulated the key falsifiability criterion: the effective rank of the per-pair covariance $\mathcal{C}_c$ in $\text{End}(V_\rho)$ should equal $d_\rho^2 = 4$ for the spin-$\frac{1}{2}$ candidate (Criterion 5.4 of [5]). Paper O27 [6] proved unconditionally that any admissible morphism $\Phi_{q,\rho} : V_q \to V_\rho$ satisfying the three structural constraints (involution equivariance, quadratic compatibility, naturality) factors uniquely through $\mathfrak{su}(2)$, closing the implication chain

$$\text{emergence} \Rightarrow \text{non-injectivity} \Rightarrow \text{pair structure} \Rightarrow \text{quadratic form} \Rightarrow \mathfrak{su}(2).$$

Paper O28 [7] performed the proxy computation of Criterion 5.4 in $\text{End}(H_{\text{eff}})$ rather than in $\text{End}(V_\rho)$, using the per-block admissible projections $\pi_c(v) \in H_{\text{eff}} = \mathbb{C}^3$ stored in the Q5a-O5 checkpoints. The result was $r_{\text{eff}} = 3$ in $\text{End}(H_{\text{eff}})$ with eigenvalue structure $[1 : \frac{1}{2} : \frac{1}{2}]$, invariant across all pairs and all $q \in \{29, 61, 101, 151\}$. O28 identified two open directions: (a) the structural origin of the $[1 : \frac{1}{2} : \frac{1}{2}]$ eigenvalue structure, and (b) the correct computation of Criterion 5.4 in $\text{End}(V_\rho)$ using the full morphism $\Phi_{q,\rho}$ of O27.

## 1.2 Scope of O29

The present paper addresses both open directions. Section 3 proves that $r_{\text{eff}} = 3$ in $\text{End}(H_{\text{eff}})$ is a structural consequence of the anti-linear pair constraint forcing the outer products $M_j$ into the complex symmetric subspace $\text{Sym}(V_\rho, \mathbb{C})$, and identifies $V_\rho \subset H_{\text{eff}}$ as the two-dimensional subspace spanned by the trajectory. Section 4 shows that direct restriction to $\text{End}(V_\rho)$ cannot yield $r_{\text{eff}} = 4$: the anti-linear constraint persists, keeping the trajectory in $\text{Sym}(2, \mathbb{C}) \subset \text{End}(\mathbb{C}^2)$. Section 5 analyses how the full admissible morphism $\Phi_{q,\rho}$ of O27 transforms the anti-linear pair constraint into charge conjugation, which is a complex-linear involution in $\text{End}(\mathbb{C}^2)$ that does not reduce the accessible dimension below 4. Section 6 gives the numerical protocol. Section 7 reports the computation from the Q5a-O5 checkpoints.

# 2 Setup and Notation

We follow the notation of O25 [4] and O28 [7]. Let $q$ be an odd prime, $G_q = \text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, and $\text{Cay}(G_q, S_q)$ its Cayley graph with generating set $S_q = \{X^{\pm 1}, Y^{\pm 1}\}$. The admissible projection $\pi_c : V_q \to H_{\text{eff}} = \mathbb{C}^3$ maps the full Weil representation space to the three-dimensional admissible subspace. The BFS fingerprint vectors at shell depth $n$ and character $c$ are denoted $v_c^{(n)} \in V_q$, and $w_j := \pi_c(v_j) \in H_{\text{eff}}$ abbreviates their admissible projections (where $j$ runs over the fitting window $[n_0, n_1]$ used in O12 [1]). The Born–Infeld parity involution of O18 [2] gives $\rho_{q-c} = \bar{\rho}_c$, which implies

$$\pi_{q-c}(v_{q-c}^{(n)}) = \overline{\pi_c(v_c^{(n)})} =: \bar{w}_j \tag{1}$$

in the canonical basis of $H_{\mathrm{eff}}$ determined by the quaternionic identification of O23 [3]. The per-pair covariance is

$$\mathcal{C}_c = \frac{1}{N} \sum_{j=n_0}^{n_1} \mathrm{vec}(M_j)\,\mathrm{vec}(M_j)^\dagger, \qquad M_j := w_j \otimes \pi_{q-c}(v_j)^* = w_j \otimes \bar{w}_j^* = w_j w_j^\top, \tag{2}$$

where the last equality uses (1). The effective rank is $r_{\mathrm{eff}} = \mathrm{rank}(\mathcal{C}_c)$ (number of eigenvalues above the threshold $10^{-2}\lambda_1$, matching the criterion of O28).

# 3 Structural Origin of $r_{\mathrm{eff}} = 3$ in $\mathrm{End}(H_{\mathrm{eff}})$

**Proposition 3.1** (Pair constraint forces symmetric outer products)**.** *Under the Born–Infeld parity constraint* (1)*, each outer product satisfies $M_j = w_j w_j^\top \in \mathrm{Sym}(3, \mathbb{C})$ (complex symmetric matrices), independently of the specific values $w_j \in H_{\mathrm{eff}}$.*

*Proof.* Direct computation: $(M_j)_{ab} = (w_j)_a (w_j)_b = (M_j)_{ba}$. □

**Proposition 3.2** (Trajectory dimension from rank equation)**.** *Let $V = \mathrm{span}_{\mathbb{C}}\{w_j : j \in [n_0, n_1]\} \subset H_{\mathrm{eff}}$ be the complex subspace spanned by the trajectory, with $\dim_{\mathbb{C}} V = k$. If $M_j = w_j w_j^\top$ and the trajectory is generic within $V$, then*

$$r_{\mathrm{eff}} = \frac{k(k+1)}{2}. \tag{3}$$

*Proof.* The set $\{ww^\top : w \in V\}$ spans $\mathrm{Sym}(V, \mathbb{C})$, the complex symmetric endomorphisms supported on $V \subset \mathbb{C}^3$. A generic trajectory within $V$ spans all of $\mathrm{Sym}(V, \mathbb{C})$, which has complex dimension $k(k+1)/2$. The eigenvalues of $\mathcal{C}_c$ above threshold are therefore $k(k+1)/2$ in number, giving $r_{\mathrm{eff}} = k(k+1)/2$. □

**Corollary 3.3** (Identification of $V_\rho$)**.** *The observed value $r_{\mathrm{eff}} = 3$ from O28 implies $k = 2$ via* (3)*. Therefore the BFS trajectory $\{w_j\}$ is effectively two-dimensional:*

$$V_\rho := \mathrm{span}_{\mathbb{C}}\{w_j : j \in [n_0, n_1]\} \subset H_{\mathrm{eff}}, \qquad \dim_{\mathbb{C}} V_\rho = 2. \tag{4}$$

*Remark* 3.4 (Structural origin of $\lambda_2 = \lambda_3$). The exact degeneracy $\lambda_2 = \lambda_3$ has a representation-theoretic explanation. With $V_\rho \cong \mathbb{C}^2$ and $w_j = (a_j, b_j)^\top$ in an orthonormal basis of $V_\rho$, the outer product $w_j w_j^\top = a_j^2 e_1 e_1^\top + a_j b_j (e_1 e_2^\top + e_2 e_1^\top) + b_j^2 e_2 e_2^\top$ lies in the three-dimensional space $\mathrm{Sym}(V_\rho, \mathbb{C}) = \mathrm{span}\{e_1 e_1^\top,\ e_1 e_2^\top + e_2 e_1^\top,\ e_2 e_2^\top\}$. The two off-diagonal modes (real and imaginary parts of $a_j b_j$) are equienergetic whenever the BFS trajectory is isotropic in $V_\rho$: $\langle |a_j b_j|^2 \rangle \cos 2\phi = 0$ for the relative phase $\phi = \arg(a_j b_j)$. Under the BFS shell dynamics, isotropy holds to high precision by the rotational symmetry of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, giving $\lambda_2 = \lambda_3$ as a structural property of the group, not a numerical coincidence.

*Remark* 3.5 (Eigenspace decomposition). The three eigenvectors of $\mathcal{C}_c$ are: (i) $f_1$, associated with $\lambda_1$, lying in $\mathrm{Sym}(V_\rho, \mathbb{C})$ and corresponding to the "radial" combination $a_j^2 e_1 e_1^\top + b_j^2 e_2 e_2^\top$ (diagonal in $V_\rho$); and (ii) $f_2, f_3$ associated with $\lambda_2 = \lambda_3$, corresponding to the real and imaginary parts of the off-diagonal mode $e_1 e_2^\top + e_2 e_1^\top$. The eigenvalue ratio $\lambda_1 : \lambda_2 = 1 : \frac{1}{2}$ reflects the relative statistical weight of diagonal versus off-diagonal components along the trajectory.

# 4 Direct Restriction to $\mathrm{End}(V_\rho)$

We now show that restricting $\mathcal{C}_c$ to $\mathrm{End}(V_\rho) = \mathrm{End}(\mathbb{C}^2)$ by projecting the trajectory onto $V_\rho$ does not raise the effective rank to 4.

Let $P : H_{\text{eff}} \to V_\rho$ denote the orthogonal projector onto $V_\rho$, and define the restricted outer products

$$\tilde{M}_j := (Pw_j) \otimes (P\bar{w}_j)^* = \tilde{w}_j\, \tilde{w}_j^\top, \qquad \tilde{w}_j := Pw_j \in \mathbb{C}^2. \tag{5}$$

Since $Pw_j = w_j$ for $w_j \in V_\rho$ (the trajectory already lies in $V_\rho$ by Corollary 3.3), we have $\tilde{M}_j = M_j|_{V_\rho} = \tilde{w}_j \tilde{w}_j^\top$.

**Proposition 4.1** (Direct restriction preserves rank 3)**.** *The covariance of* $\{\mathrm{vec}(\tilde{M}_j)\}$ *in* $\mathrm{End}(\mathbb{C}^2)$ *has effective rank* $r_{\text{eff}} = 3$ *for a generic trajectory* $\{\tilde{w}_j\}$ *spanning* $\mathbb{C}^2$*.*

*Proof.* $\tilde{M}_j = \tilde{w}_j \tilde{w}_j^\top \in \mathrm{Sym}(2, \mathbb{C})$ by the same argument as Proposition 3.1. $\mathrm{Sym}(2, \mathbb{C})$ has complex dimension $2(2+1)/2 = 3$ and is a proper subspace of $\mathrm{End}(\mathbb{C}^2)$ (complex dimension 4). A generic trajectory spanning $\mathbb{C}^2$ fills $\mathrm{Sym}(2, \mathbb{C})$ completely, giving $r_{\text{eff}} = 3 < 4 = d_\rho^2$. $\square$

This result establishes that direct restriction cannot test Criterion 5.4: the spin-$\frac{1}{2}$ prediction $d_\rho^2 = 4$ is not accessible via the projection approach. The missing fourth dimension corresponds to the anti-Hermitian subspace $\mathrm{AHerm}(2, \mathbb{C}) = \{A \in \mathrm{End}(\mathbb{C}^2) : A = -A^\dagger\} \cap \mathrm{Sym}(2, \mathbb{C})^\perp$; direct restriction keeps the trajectory in the symmetric subspace and does not probe this direction.

# 5 Rank Restoration via the Full Admissible Morphism

The correct operationalisation of O26 Criterion 5.4 requires the full admissible morphism $\Phi_{q,\rho} = \rho \circ \iota \circ \pi$ of O27 [6], where: $\pi : V_q \to H_{\text{eff}}$ is the admissible quotient map, $\iota : H_{\text{eff}} \xrightarrow{\sim} \mathfrak{su}(2)$ is the canonical quaternionic identification of O23 [3], and $\rho : \mathfrak{su}(2) \to \mathrm{End}(V_\rho)$ is the spin-$\frac{1}{2}$ Lie algebra action.

## 5.1 The pair constraint under the full morphism

The O27 morphism maps $\Phi_{q,\rho}(v_c^{(n)}) \in \mathrm{End}(V_\rho)$ (an operator on $V_\rho$, not a vector in $V_\rho$). To obtain vectors in $V_\rho = \mathbb{C}^2$, we choose a fixed highest-weight state $|\psi_0\rangle \in V_\rho$ determined by the admissibility thread $\mathrm{Q}_8 \subset 2I \subset \mathrm{SU}(2)$ (see O26 Section 6.3 [5]), and define

$$\hat{u}_j := \rho(\iota(w_j))\, |\psi_0\rangle \in V_\rho = \mathbb{C}^2. \tag{6}$$

The pair constraint (1) maps $w_j \mapsto \bar{w}_j$, which under $\iota$ (a real isomorphism $H_{\text{eff}} \to \mathfrak{su}(2)$) becomes complex conjugation $X \mapsto X^*$ in $\mathfrak{su}(2)$.

**Lemma 5.1** (Charge conjugation)**.** *In the spin-*$\frac{1}{2}$ *representation* $\rho$ *of* $\mathfrak{su}(2)$*, complex conjugation acts by*

$$\rho(X^*) = C\, \rho(X)\, C^{-1}, \qquad C := i\sigma_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \tag{7}$$

*where* $\sigma_2$ *is the second Pauli matrix. This is the standard charge-conjugation matrix for the spin-*$\frac{1}{2}$ *representation; it satisfies* $C^\dagger = C^{-1}$*,* $C^2 = -I$*.*

*Proof.* The spin-$\frac{1}{2}$ matrices satisfy $\rho(e_k)^* = -\rho(e_k)^\top$ for the standard generators $e_k = i\sigma_k/2$. The identity $C\sigma_k^\top C^{-1} = -\sigma_k$ (equivalently $C\sigma_k^* C^{-1} = \sigma_k$) is a standard result of representation theory and is verified by direct computation. $\square$

Under the pair constraint, the morphism therefore sends

$$\hat{u}_j \mapsto \hat{u}_j^{(q-c)} := \rho(\iota(\bar{w}_j))\,|\psi_0\rangle = C\,\rho(\iota(w_j))\,C^{-1}|\psi_0\rangle = C\,\rho(\iota(w_j))\,|\psi_0'\rangle, \tag{8}$$

where $|\psi_0'\rangle := C^{-1}|\psi_0\rangle$ is a second reference state. The outer product under the full morphism is

$$\hat{M}_j := \hat{u}_j \otimes (\hat{u}_j^{(q-c)})^* = \rho(X_j)|\psi_0\rangle \otimes \big(C\,\rho(X_j)|\psi_0'\rangle\big)^*, \qquad X_j := \iota(w_j). \tag{9}$$

This is a rank-one operator in $\mathrm{End}(\mathbb{C}^2)$ of the form $|u\rangle\langle\tilde{u}|$ with $|u\rangle = \rho(X_j)|\psi_0\rangle$ and $\langle\tilde{u}| = |\psi_0'\rangle^\dagger \rho(X_j)^\dagger C^\dagger$.

## 5.2 The pair constraint is complex-linear in $\mathrm{End}(\mathbb{C}^2)$

**Proposition 5.2** (Rank restoration)**.** *Under the full admissible morphism, the outer products $\hat{M}_j$ are not confined to* $\mathrm{Sym}(2,\mathbb{C})$*. For a generic trajectory $\{X_j = \iota(w_j)\} \subset \mathfrak{su}(2)$ spanning the three generators, the set $\{\hat{M}_j\}$ spans all four dimensions of* $\mathrm{End}(\mathbb{C}^2)$ *[H2].*

*Structural argument (H2).* The anti-linear pair constraint $w_j \mapsto \bar{w}_j$ becomes, after the two successive real-linear maps $\iota$ and $\rho(\cdot)|\psi_0\rangle$, the complex-linear relation $\hat{u}_j \mapsto C\,\rho(X_j)\,|\psi_0'\rangle$. Since $C \neq \pm I$ (indeed $C^2 = -I$), the map $\hat{u}_j \mapsto \hat{u}_j^{(q-c)}$ is a non-scalar complex-linear transformation. The outer product $\hat{M}_j = |u_j\rangle\langle\tilde{u}_j|$ with $u_j$ and $\tilde{u}_j$ in general complex-linearly related positions therefore does not satisfy $\hat{M}_j = \hat{M}_j^\top$ in general. As $X_j$ varies generically over the trajectory, the rank-one operators $|u_j\rangle\langle\tilde{u}_j|$ span $\mathrm{End}(\mathbb{C}^2)$ (four dimensions), as is the case for a generic pair of curves $u_j, \tilde{u}_j$ in $\mathbb{C}^2$. $\square$

*Remark* 5.3 (Status of Proposition 5.2)*.* Proposition 5.2 carries status [H2]: the structural argument is tight (the charge-conjugation identity is proved; the genericity assumption for the trajectory follows from the BFS dynamics filling $\mathfrak{su}(2)$ in three independent directions, consistent with $\Sigma_c(n_3) = 3$ from O23 [3] and the rank-3 result of O28 [7]). The numerical computation of Section 7 provides the required empirical confirmation.

## 5.3 Summary of the two approaches

Table 1 contrasts the two computation strategies.

| Approach | Outer product | Constraint in $\mathrm{End}(\mathbb{C}^2)$ | Predicted $r_{\mathrm{eff}}$ |
|---|---|---|---|
| Direct restriction (Sec. 4) | $\tilde{w}_j\tilde{w}_j^\top$ | anti-linear ($\subset \mathrm{Sym}(2,\mathbb{C})$) | 3 |
| Full morphism (Sec. 5) | $\hat{u}_j \otimes (\hat{u}_j^{(q-c)})^*$ | complex-linear ($C = i\sigma_2$) | 4 [H2] |

Table 1: The two approaches to computing $r_{\mathrm{eff}}$ in $\mathrm{End}(\mathbb{C}^2)$. Only the full morphism (Approach B) is the correct operationalisation of O26 Criterion 5.4.

# 6 Numerical Protocol

## 6.1 Data source

The computation uses the Q5a-O5 checkpoints (one `.npz` file per prime $q$), which store the arrays `V_n` (the per-shell admissible projections $\pi_c(v^{(n)}) \in \mathbb{C}^3$) and `all_sigma` (the pair observables). No additional BFS computation is required.

## 6.2 Step A: identification of $V_\rho$ and the su(2) identification $\iota$

For each conjugate pair $(c, q-c)$, collect the trajectory matrix

$$W_c = \big[w_{n_0},\, w_{n_0+1},\, \ldots,\, w_{n_1}\big] \in \mathbb{C}^{3\times N}, \qquad w_j = \texttt{V\_n}[c, j]. \tag{10}$$

Compute the singular value decomposition $W_c = U\Sigma V^\dagger$. The two leading left singular vectors $\{u_1, u_2\}$ of $U$ span the candidate $V_\rho \subset \mathbb{C}^3$, corresponding to the two non-negligible singular values $\sigma_1 \geq \sigma_2 \gg \sigma_3 \approx 0$. The projector onto $V_\rho$ is $P_V = u_1 u_1^\dagger + u_2 u_2^\dagger$.

The identification $\iota : H_{\text{eff}} \to \mathfrak{su}(2)$ is fixed as follows. Choose an orthonormal basis $\{e_1, e_2, e_3\}$ of $\mathbb{C}^3$ aligned with the three singular vectors of a reference pair at each prime $q$. The map $\iota$ sends $e_k$ to the $k$-th generator of $\mathfrak{su}(2)$: $\iota(e_k) = i\sigma_k/2$ (canonical identification via O23 [3]). The spin-$\frac{1}{2}$ representation is $\rho(i\sigma_k/2) = i\sigma_k/2 \in \text{End}(\mathbb{C}^2)$.

## 6.3 Step B: direct restriction (Approach A, falsifiability check)

For each $j \in [n_0, n_1]$, set $\tilde{w}_j = P_V w_j \in \mathbb{C}^2$ (two-component vector in the basis $\{u_1, u_2\}$). Form $\tilde{M}_j = \tilde{w}_j \tilde{w}_j^\top \in \text{End}(\mathbb{C}^2)$. Compute

$$\tilde{\mathcal{C}}_c^{(A)} = \frac{1}{N} \sum_{j=n_0}^{n_1} \text{vec}(\tilde{M}_j)\, \text{vec}(\tilde{M}_j)^\dagger \in \text{End}(\mathbb{C}^4). \tag{11}$$

Record $r_{\text{eff}}^{(A)} = \text{rank}(\tilde{\mathcal{C}}_c^{(A)})$ (threshold $10^{-2}\lambda_1$). Proposition 4.1 predicts $r_{\text{eff}}^{(A)} = 3$ for all pairs.

## 6.4 Step C: full morphism (Approach B, Criterion 5.4)

For each $j$, compute the $\mathfrak{su}(2)$ element $X_j = \iota(w_j) = \sum_{k=1}^{3} (w_j)_k\, (i\sigma_k/2) \in \mathfrak{su}(2)$ and the representation matrix $A_j = \rho(X_j) \in \text{End}(\mathbb{C}^2)$. Apply to the reference state $|\psi_0\rangle = (1, 0)^\top$ and the conjugated reference state $|\psi_0'\rangle = C^{-1}|\psi_0\rangle = (0, 1)^\top$ (with $C = i\sigma_2$):

$$\hat{u}_j = A_j\, |\psi_0\rangle \in \mathbb{C}^2, \qquad \hat{u}_j^{(q-c)} = C\, A_j\, |\psi_0'\rangle \in \mathbb{C}^2. \tag{12}$$

Form $\hat{M}_j = \hat{u}_j \otimes (\hat{u}_j^{(q-c)})^* \in \text{End}(\mathbb{C}^2)$ and compute

$$\tilde{\mathcal{C}}_c^{(B)} = \frac{1}{N} \sum_{j=n_0}^{n_1} \text{vec}(\hat{M}_j)\, \text{vec}(\hat{M}_j)^\dagger \in \text{End}(\mathbb{C}^4). \tag{13}$$

Record $r_{\text{eff}}^{(B)} = \text{rank}(\tilde{\mathcal{C}}_c^{(B)})$ (threshold $10^{-2}\lambda_1$). Proposition 5.2 predicts $r_{\text{eff}}^{(B)} = 4$ [H2].

| Outcome | Conclusion | Status |
|---|---|---|
| $r_{\text{eff}}^{(A)} = 3$ (all pairs) | Proposition 4.1 confirmed | proved prec |
| $r_{\text{eff}}^{(B)} = 4$ (all pairs) | spin-$\frac{1}{2}$ sector validated; Level III of O26 | [H2] confirm |
| $r_{\text{eff}}^{(B)} = 3$ (all pairs) | pair constraint not lifted by the morphism; sector revision needed | |
| $r_{\text{eff}}^{(B)}$ pair-dependent | Level II at most; canonical identification fails | |

Table 2: Decision table for the Approach A and Approach B computations.

### 6.5 Falsifiability table

## 7 Results

The computation was carried out from the Q5a-O5 checkpoints for $q \in \{61, 101, 151\}$. (At $q = 29$, only 5/14 pairs have non-empty trajectory arrays; the structure analysis holds but with reduced statistics.)

### 7.1 Approach A: direct restriction

For every conjugate pair $(c, q-c)$ at all three primes, the effective rank of $\tilde{\mathcal{C}}_c^{(A)}$ equals 3 (threshold $10^{-2}\lambda_1$), with eigenvalue structure $[\tilde{\lambda}_1 : \tilde{\lambda}_2 : \tilde{\lambda}_3 : 0] = [1 : \frac{1}{2} : \frac{1}{2} : 0]$. This is identical to the O28 result in $\mathrm{End}(H_{\text{eff}})$, confirming Proposition 4.1: the direct restriction preserves the rank-3 constraint. Inter-pair variance is negligible ($< 1\%$) at all primes.

### 7.2 Approach B: full admissible morphism

For every conjugate pair $(c, q-c)$ at $q \in \{61, 101, 151\}$, the effective rank of $\tilde{\mathcal{C}}_c^{(B)}$ equals 4 (threshold $10^{-2}\lambda_1$), with all four eigenvalues above threshold. The normalised eigenvalue structure is $[\hat{\lambda}_1 : \hat{\lambda}_2 : \hat{\lambda}_3 : \hat{\lambda}_4] \approx [1 : a : b : c]$ with $a, b, c > 0$ (no exact degeneracy), depending weakly on the prime $q$ and on the specific pair $(c, q-c)$. Inter-pair variance is small but non-zero, reflecting the pair-dependence of the reference state alignment.

| Prime $q$ | $r_{\text{eff}}^{(A)}$ (all pairs) | $r_{\text{eff}}^{(B)}$ (all pairs) |
|---|---|---|
| 61 | 3 | 4 |
| 101 | 3 | 4 |
| 151 | 3 | 4 |

Table 3: Effective rank under Approach A (direct restriction) and Approach B (full morphism) for all conjugate pairs at each prime. The rank-4 result of Approach B confirms the spin-$\frac{1}{2}$ sector prediction $d_\rho^2 = 4$ [H2].

## 8 Discussion

### 8.1 What O29 establishes

Three results are established.

First, the rank-3 result of O28 [7] in $\mathrm{End}(H_{\text{eff}})$ is analytically explained: it is a structural consequence of the pair constraint forcing $M_j \in \mathrm{Sym}(V_\rho, \mathbb{C})$, with $\dim_{\mathbb{C}} V_\rho = 2$

determined by the equation $k(k+1)/2 = 3$. The exact degeneracy $\lambda_2 = \lambda_3$ is a consequence of the BFS trajectory being isotropic within $V_\rho$, itself a structural property of the group dynamics.

Second, direct restriction to $\mathrm{End}(V_\rho)$ is not the correct operationalisation of O26 Criterion 5.4 [5]: it preserves the anti-linear pair constraint and gives $r_{\text{eff}} = 3$ in $\mathrm{End}(\mathbb{C}^2)$, falling short of $d_\rho^2 = 4$. This is not a failure of the spin-$\frac{1}{2}$ identification but a structural feature of the anti-linear pair action.

Third, the full admissible morphism $\Phi_{q,\rho} = \rho \circ \iota \circ \pi$ of O27 [6] transforms the anti-linear pair constraint into charge conjugation $C = i\sigma_2$ in $\mathrm{End}(\mathbb{C}^2)$, a complex-linear involution that does not restrict the trajectory to a proper subspace of $\mathrm{End}(\mathbb{C}^2)$. Under Approach B, $r_{\text{eff}} = 4$ is confirmed numerically for all pairs at $q \in \{61, 101, 151\}$, validating O26 Level III and confirming the spin-$\frac{1}{2}$ sector candidate.

## 8.2 Completion of the O26 hierarchy

The O26 decision table [5] (Section 5.6 therein) is now resolved: Tests 1–5 pass under Approach B ($r_{\text{eff}}^{(B)} = 4$, universally across pairs). The identification $\mathrm{End}(V_\rho) \cong \mathbb{C}^{2\times 2}$ as the minimal ambient space for admissible spectral dynamics is confirmed, completing Level III of the O26 dictionary.

## 8.3 Interpretation of charge conjugation in the framework

The charge-conjugation matrix $C = i\sigma_2$ has a structural role beyond its technical function in the computation. In the Cosmochrony framework [9], the Born–Infeld parity involution $c \leftrightarrow q-c$ (O18 [2]) is the discrete realisation of the substrate symmetry $\chi \mapsto -\chi$. Its image in $\mathrm{End}(V_\rho)$ under the spin-$\frac{1}{2}$ representation is $C = i\sigma_2$, the standard charge-conjugation operator of the spin-$\frac{1}{2}$ sector. The fact that this operator is non-scalar ($C^2 = -I$) is precisely what allows the outer products $\hat{M}_j$ to escape the symmetric subspace: a scalar phase ($C = e^{i\theta} I$) would leave $\hat{M}_j \in \mathrm{Sym}(2, \mathbb{C})$. The non-scalar nature of $C$ is in turn a structural consequence of $V_\rho$ being a non-trivial irreducible SU(2)-module (the trivial representation has $C = I$, the spin-1 representation is real and has no such $C$). The spin-$\frac{1}{2}$ sector is therefore the unique sector for which the Born–Infeld parity involution lifts to a non-scalar unitary in $\mathrm{End}(V_\rho)$, enabling $r_{\text{eff}} = d_\rho^2 = 4$.

## 8.4 What remains open

The following directions are open after O29:

- Analytical determination of the normalised eigenvalue structure $[\hat{\lambda}_1 : \hat{\lambda}_2 : \hat{\lambda}_3 : \hat{\lambda}_4]$ under Approach B and its dependence on $q$.
- Extension of the Approach B computation to $q \in \{211, 307, 401\}$ to confirm the universality of $r_{\text{eff}}^{(B)} = 4$ across the full tested range.
- Formal proof that the genericity condition in Proposition 5.2 is satisfied by the BFS trajectory for all odd primes $q$ (currently supported by the three-direction spanning established numerically via O23 and O28).

# 9 Conclusion

O29 resolves the two open directions of O28 [7]. On the structural side, the rank-3 result in $\mathrm{End}(H_{\mathrm{eff}})$ is explained by the anti-linear pair constraint forcing $M_j \in \mathrm{Sym}(V_\rho, \mathbb{C})$ with $\dim_{\mathbb{C}} V_\rho = 2$. On the numerical side, the full admissible morphism $\Phi_{q,\rho}$ of O27 [6] transforms this anti-linear constraint into the complex-linear charge-conjugation operator $C = i\sigma_2$, lifting the symmetric restriction and restoring $r_{\mathrm{eff}} = 4 = d_\rho^2$ in $\mathrm{End}(\mathbb{C}^2)$. This confirms the spin-$\frac{1}{2}$ sector candidate [H2], completes O26 Level III [5], and closes the representation-theoretic component of the O-series. The full implication chain

$$\text{emergence} \Rightarrow \text{non-injectivity [8]} \Rightarrow \text{pair structure [2]} \Rightarrow \mathfrak{su}(2) \text{ [6]} \Rightarrow \text{spin-}\tfrac{1}{2} \text{ [O29]}$$

is thereby closed, with each arrow a proved or numerically confirmed result.

*Acknowledgements.* The author acknowledges the use of large language models as a supportive tool for refining language, structure, and internal consistency during the development of this manuscript.

# References

[1] J. Beau. Exact Weil-Block Projective Capacity on Heisenberg Graphs: Resolving the Final Obstruction to $\delta$ Extraction. Preprint, 2026. O12 paper. doi:10.5281/zenodo.19194798

[2] J. Beau. Minimal Fibre Structure of the Non-Injective Projection from Born–Infeld Indiscernability: Derivation of the Parity Involution. Preprint, 2026. O18 paper. doi:10.5281/zenodo.19321413

[3] J. Beau. Three Stable Directions from Quaternionic Minimality: Derivation of $\Sigma_c(n_3) = 3$ from Born–Infeld Fibre Admissibility. Preprint, 2026. O23 paper. doi:10.5281/zenodo.19375136

[4] J. Beau. Systematic Pair-Level Campaign for $\delta_{\mathrm{pair}}$: Convergence, Inter-Pair Concentration, and Normalization Structure. Preprint, 2026. O25 paper. doi:10.5281/zenodo.19462132

[5] J. Beau. Quadratic Completion of Admissible Spectral Pairs via Binary-Icosahedral Representation. Preprint, 2026. O26 paper. doi:10.5281/zenodo.19488845

[6] J. Beau. Quaternionic Rigidity of Admissible Morphisms: Every Admissible $\Phi_{q,\rho}$ Necessarily Factors through $\mathfrak{su}(2)$. Preprint, 2026. O27 paper. doi:10.5281/zenodo.19489296

[7] J. Beau. Asymptotic Calibration of the BFS Window and Effective Dimension of the Admissible Trajectory. Preprint, 2026. O28 paper. doi:10.5281/zenodo.19767801

[8] J. Beau. Non-Injectivity as a Structural Necessity of Genuine Emergence: A No-Go Theorem for Faithful Emergent Descriptions. Preprint, 2026. Cosmochrony companion paper "ENI"/L. doi:10.5281/zenodo.19391746

[9] J. Beau. Cosmochrony: A Non-Injective Projection Theory of Emergent Physics. Preprint, 2026. White paper. doi:10.5281/zenodo.17957509